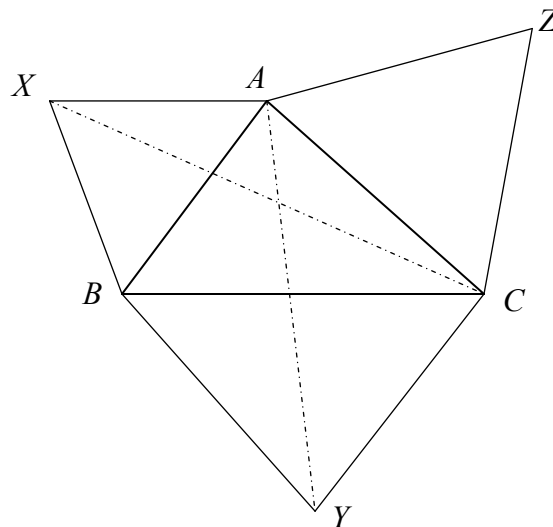


Question 1.**Marks**

- (a) Rationalise the denominator of: $\frac{4\sqrt{2}-3}{\sqrt{2}+5}$. **2**
- (b) Simplify: $\frac{2}{x^2-4} - \frac{3}{x+2} + \frac{5}{x-2}$. **2**
- (c) Evaluate: $\lim_{x \rightarrow 0} \frac{\sin\left(\frac{x}{5}\right)}{2x}$. **2**
- (d) Solve for t : $(\pi - t)(t - 3) < 0$. **2**
- (e) Differentiate the following with respect to x .
- (i) ex^2 . **1**
- (ii) 7^x . **1**
- (iii) $\frac{1}{\sqrt{1+x^2}}$. **2**
- (iv) $\ln\left(\frac{x+1}{\sqrt{\sin 2x-1}}\right)$. **3**

Question 2.	[Start a New page]	Marks
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- | | | |
|------|---|----------|
| (a) | If α and β are the roots of the equation $x^2 - 2x - 4 = 0$, | |
| (i) | Show that $\alpha^2 + \beta^2 = 12$. | 2 |
| (ii) | Hence, or otherwise find $\alpha^3 + \beta^3$. | 1 |
| | | |
| (b) | Solve for x : $ x - 2 < 3 - x $. | 3 |
| | | |
| (c) | Given $f(x) = 3x \ln^2 x$, find $f'(x)$. | 3 |
| | | |
| (d) | Shade the region on the number plane where: $y > \sqrt{x}$. | 2 |
| | | |
| (e) | Equilateral triangles ABX , CBY and CAZ are constructed externally on the sides AB , BC and CA of $\triangle ABC$ respectively, as shown below. | |



Not to scale

- | | | |
|------|--|----------|
| (i) | Copy the diagram onto your writing paper and
Prove that: $\triangle BXC \equiv \triangle BAY$. | 2 |
| | | |
| (ii) | Hence, or otherwise, explain why $BZ = AY = CX$. | 2 |

Question 3.

[Start a New Page]

Marks

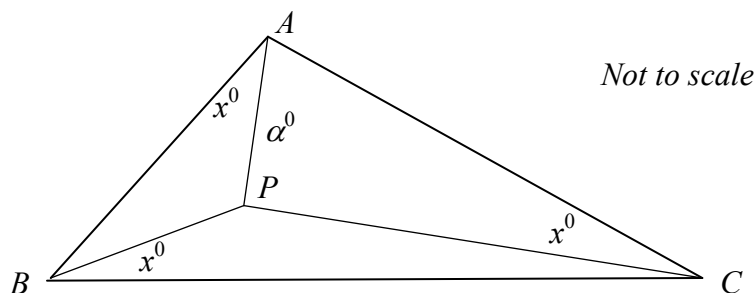
(a) Solve for x : $\sqrt{3} \tan(x - \frac{\pi}{3}) = 1$, for $0 \leq x \leq 2\pi$. **2**

(b) (i) Sketch the graph of $y = 1 - \cos \pi x$ over the domain $0 \leq x \leq 2$. **2**

(ii) Find the equation of the tangent to $y = 1 - \cos \pi x$ at $x = \frac{1}{3}$. **4**

(c) Prove that: $\frac{\sin \theta}{\cos \theta + \sin \theta} + \frac{\sin \theta}{\cos \theta - \sin \theta} \equiv \tan 2\theta$. **2**

(d) In $\triangle ABC$, $\angle CAB = \alpha^\circ$, $\angle ABC = \beta^\circ$ and $\angle BCA = \gamma^\circ$.
The point P is chosen internally in $\triangle ABC$ so that:
 $\angle PAB = \angle PBC = \angle PCA = x^\circ$ as shown in the diagram.



(i) Show that: $\sin(\beta - x)^\circ = \frac{PA}{PB} \sin x^\circ$. **1**

(ii) Hence show that: $\sin(\alpha - x)^\circ \sin(\beta - x)^\circ \sin(\gamma - x)^\circ = \sin^3 x^\circ$. **2**

(iii) Using the identity: $\cot p - \cot q = \frac{\sin(q - p)}{\sin p \sin q}$, deduce that: **2**

$$(\cot x^\circ - \cot \alpha^\circ)(\cot x^\circ - \cot \beta^\circ)(\cot x^\circ - \cot \gamma^\circ) = \operatorname{cosec} \alpha^\circ \operatorname{cosec} \beta^\circ \operatorname{cosec} \gamma^\circ.$$

Question 4.	[Start a New Page]	Marks
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(a)	Consider the statement:	2
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' For a point of inflection to exist at $x = c$ on a continuous curve $y = f(x)$, then $f'(c) = f''(c) = 0$ only '.

Discuss the truthfulness of this statement, giving reasons.

(b)	Consider the curve: $y = (x + \frac{1}{x})^2 - 5(x + \frac{1}{x}) + 6\frac{1}{4}$.	
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(i)	Determine the x -intercepts.	2
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(ii)	Show that: $\frac{dy}{dx} = \left(1 - \frac{1}{x^2}\right)\left(2x - 5 + \frac{2}{x}\right)$.	2
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(iii)	Hence determine the stationary points of this curve.	3
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(iv)	Determine the nature of these stationary points.	3
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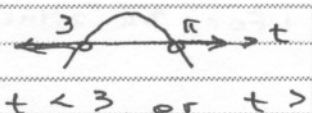
(v)	Hence sketch the curve $y = (x + \frac{1}{x})^2 - 5(x + \frac{1}{x}) + 6\frac{1}{4}$. [show all essential detail].	3
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THE END

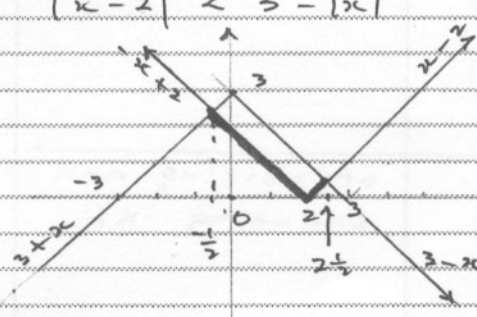
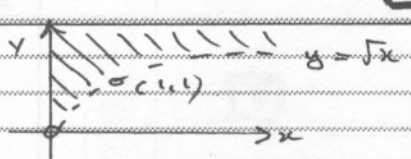


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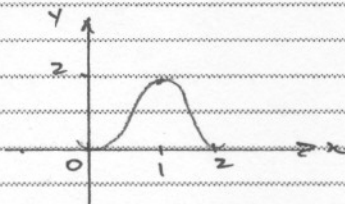
MATHEMATICS Extension 1: Question 1

Suggested Solutions	Marks	Marker's Comments
Q1(a) $\frac{4\sqrt{2}-3}{\sqrt{2}+5} \times \frac{(5-\sqrt{2})}{(5-\sqrt{2})} = \frac{20\sqrt{2}-8-15+3\sqrt{2}}{25-2}$ $= \frac{23\sqrt{2}-23}{23} = \sqrt{2}-1$	1 1	$\times \frac{\sqrt{2}-5}{\sqrt{2}-5}$
(b) $\frac{2}{x^2-4} - \frac{3}{x+2} + \frac{5}{x-2} = \frac{2}{(x+2)(x-2)} - \frac{3}{x+2} + \frac{5}{x-2}$ $= \frac{2 - 3(x-2) + 5(x+2)}{(x+2)(x-2)}$ $= \frac{2 - 3x + 6 + 5x + 10}{(x+2)(x-2)} = \frac{2x+18}{(x+2)(x-2)}$	1 1	
(c) METHOD 1 $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{5}}{2x} = \lim_{x \rightarrow 0} \frac{\sin \frac{x}{5}}{2(\frac{x}{5}) \cdot 5}$ $= \frac{1}{10} \times 1 = \frac{1}{10}$	1 1	organising and showing $\times 1$
(d) $(\pi-t)(t-3) < 0$ $\pi > 3$  $\therefore t < 3 \text{ or } t > \pi$	1, 1	1 for '$t < 3$' 1 for '$\text{or } t > \pi$' All for $3 < t < \pi$
(e) (i) let $y = e^{2x}$ $y' = 2e^{2x}$	1	
(ii) $y = 7^x = e^{x \ln 7}$ $y' = \ln 7 \cdot 7^x$	1	
(iii) $y = \frac{1}{\sqrt{1+x^2}} = (1+x^2)^{-\frac{1}{2}}$ $y' = -\frac{1}{2}(1+x^2)^{-\frac{3}{2}} \times 2x$ $= -x(1+x^2)^{-\frac{3}{2}}$	1 1	
(iv) $y = \ln \frac{x+1}{\sqrt{\sin 2x-1}} = \ln(x+1) - \frac{1}{2} \ln(\sin 2x-1)$ $y' = \frac{1}{x+1} - \frac{1}{2} \frac{\cos 2x}{\sin 2x-1}$ $= \frac{1}{x+1} - \frac{\cos 2x}{2(\sin 2x-1)}$	1, 1	

MATHEMATICS Extension 1 : Question 2

Suggested Solutions	Marks	Marker's Comments
Q2(a) $x^2 - 2x - 4 = 0$ [2] (i) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ $= 2^2 - 2 \times (-4)$ $= 4 + 8 = 12$ quad. (ii) $\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 + \beta^2 - \alpha\beta)$ [1] $= 2(12 - (-4))$ $= 32$	1 1 1	✓ ✓
(b) $x-2 < 3- x$  $-x+2 = 3+x$ $-1 = 2x$ $\therefore x = -\frac{1}{2}$ $3-x = x+2$ $5 = 2x$ $x = 2\frac{1}{2}$ $\therefore -\frac{1}{2} < x < 2\frac{1}{2}$ [3]	1 1 1	
(c) $f(x) = 3x \ln^2 x$ $f'(x) = 3 \ln^2 x + 3x \times 2 \ln x \times \frac{1}{x}$ $= 3 \ln^2 x + 6 \ln x$ [3]	1, 1 1	1 For $3 \ln^2 x +$ 1 For $3x \cdot 2 \ln x \cdot \frac{1}{x}$
(d) $y > \sqrt{x}$  [2]		
(e) (i) In ΔBXC and ΔBAY S 1. $BX = AB$ (given data) A 2. $\angle XBC = \angle ABY = 60^\circ + \angle ABC$ ✓ (all angles in equilateral Δ are 60°) S 3. $BC = BY$ (given data) $\therefore \Delta BXC \equiv \Delta BAY$ (SAS) [2]	1	1 For $BX = AB$ or $BC = BY$
(ii) $CX = AY$ (corresponding sides in congruent triangles are equal) 1 By similar argument $\Delta ABY \equiv \Delta ZCB$ Hence $BZ = AY = CX$ [2]	1	1 For a sound argument.

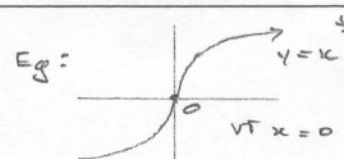
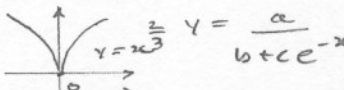
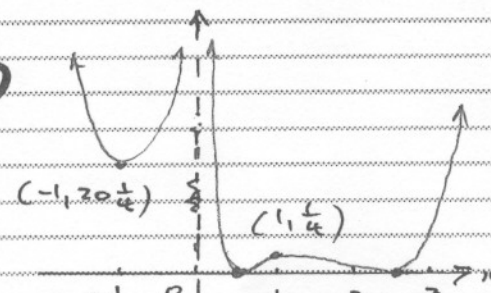
MATHEMATICS Extension 1 : Question 3

Suggested Solutions	Marks	Marker's Comments
63 (a) $\sqrt{3} \tan(x - \frac{\pi}{3}) = 1$ $\tan(x - \frac{\pi}{3}) = \frac{1}{\sqrt{3}}$ [3] $x - \frac{\pi}{3} = \frac{\pi}{6} \text{ or } \pi + \frac{\pi}{6}$ $x = \frac{3\pi}{6} \text{ or } \frac{7\pi}{6}$ $\frac{\pi}{2}, \frac{3\pi}{2}$ 1, 1		
(b) (i)  $y = 1 - \cos \pi x, 0 \leq x \leq 2$ [2]		
(ii) $y = 1 - \cos \pi x$ at $x = \frac{1}{3}, y = 1 - \cos \pi = \frac{1}{2}$ ✓ $(\frac{\pi}{3}, \frac{1}{2})$ $y' = \pi \sin \pi x$ ✓ Gradient of tangent at $x = \frac{1}{3}: m_T = \pi \times \frac{\sqrt{3}}{2}$ ✓ Equation of tangent at $x = \frac{1}{3}$ $y - \frac{1}{2} = \frac{\pi\sqrt{3}}{2} (x - \frac{1}{3})$ ✓ $y = \frac{\pi\sqrt{3}}{2} x + \frac{1}{2} - \frac{\pi\sqrt{3}}{6}$	1 1 1 1	$3\pi\sqrt{3}x - 6y + 3 - \pi\sqrt{3} = 0$
(c) $\frac{\sin \theta}{\cos \theta + \sin \theta} + \frac{\sin \theta}{\cos \theta - \sin \theta} =$ [2] $= \frac{\sin \theta \cos \theta - \sin^2 \theta + \sin \theta \cos \theta + \sin^2 \theta}{\cos^2 \theta - \sin^2 \theta}$ $= \frac{2 \sin \theta \cos \theta}{\cos^2 \theta - \sin^2 \theta} = \frac{\sin 2\theta}{\cos 2\theta}$ $= \tan 2\theta$ qed.	1 1	

MATHEMATICS Extension 1 : Question 3

Suggested Solutions	Marks	Marker's Comments
3(d) (i) In $\triangle APB$, using the sine rule $\frac{PA}{\sin(\beta-x)^\circ} = \frac{PB}{\sin x^\circ} \quad \text{or} \quad \boxed{1}$ $\frac{PA}{PB} = \frac{\sin x^\circ}{\sin(\beta-x)^\circ}$ $\therefore \sin(\beta-x)^\circ = \frac{PA \sin x^\circ}{PB}$	1	
(ii) From (i) $\sin(\alpha-x)^\circ = \frac{PC \sin x^\circ}{PA}$ and $\sin(\gamma-x)^\circ = \frac{PB \sin x^\circ}{PC}$ $\therefore \sin(\alpha-x)^\circ \sin(\beta-x)^\circ \sin(\gamma-x)^\circ$ $= \frac{PC \sin x^\circ}{PA} \times \frac{PA \sin x^\circ}{PB} \times \frac{PB \sin x^\circ}{PC}$ $= \sin^3 x^\circ \quad \text{q.e.d.}$	1	
(iii) $(\cot x^\circ - \cot \alpha^\circ)(\cot x^\circ - \cot \beta^\circ)(\cot x^\circ - \cot \gamma^\circ)$ $= \frac{\sin(\alpha-x)^\circ}{\sin \alpha^\circ \sin x^\circ} \times \frac{\sin(\beta-x)^\circ}{\sin \beta^\circ \sin x^\circ} \times \frac{\sin(\gamma-x)^\circ}{\sin \gamma^\circ \sin x^\circ}$ $= \frac{\sin^3 x^\circ}{\sin \alpha^\circ \sin \beta^\circ \sin \gamma^\circ} \quad \text{using (ii)}$ $= \frac{1}{\sin \alpha^\circ \sin \beta^\circ \sin \gamma^\circ}$ $= \operatorname{cosec} \alpha^\circ \operatorname{cosec} \beta^\circ \operatorname{cosec} \gamma^\circ \quad \text{q.e.d.}$	2	

MATHEMATICS Extension 1 : Question 4

Suggested Solutions	Marks	Marker's Comments
4(a) This is not 100% True A POI is a point where the curve changes its concavity. Eg: $y = x^3$ at $x = c = 0$ $f(0) = 0$ but $f'(0)$ and $f''(0) \neq 0$ a. V.P.OI at $c = 0$	1 1	Eg:  
(b) (i) Let $u = x + \frac{1}{x}$ $\therefore u^2 - 5u + 6\frac{1}{x} = 0$ $u = \frac{5}{2}$ $x + \frac{1}{x} = \frac{5}{2}$ $2x^2 - 5x + 2 = 0$ } $x = \frac{1}{2}$ or 2	1, 1	x-intercepts $x = \frac{1}{2}$ and 2
(ii) $y = (x + \frac{1}{x})^2 - 5(x + \frac{1}{x}) + 6\frac{1}{x}$ $\frac{dy}{dx} = 2(x + \frac{1}{x})(1 - \frac{1}{x^2}) - 5(1 - \frac{1}{x^2})$ $= (1 - \frac{1}{x^2})(2(x + \frac{1}{x}) - 5)$	1 1	
(iii) For s.p.s to occur $\frac{dy}{dx} = 0$ $\therefore (1 - \frac{1}{x^2})(2x - 5 + \frac{2}{x}) = 0$ $\therefore x = \pm 1, \frac{1}{2}$ or 2 $\therefore$ s.p.s are $(-1, 20\frac{1}{4})$, $(1, \frac{1}{4})$ $(\frac{1}{2}, 0)$ and $(2, 0)$	• 1, 1 1	
(iv) • From (i) and (iii) $(\frac{1}{2}, 0)$ and $(2, 0)$ are <u>neut</u> TP. At $(-1, 20\frac{1}{4})$ is a <u>neut</u> TP At $(1, \frac{1}{4})$ is a <u>max</u> TP	1 1 1	1 For testing at $x = -1$ 1 For testing at $x = 1$ [1st or 2nd deriv]
(v) 	3	$\frac{1}{2}$ For VA at $x = 0$ $\frac{1}{2}$ For min TP $\frac{1}{2}$ For max TP $\frac{1}{2}$ For shape

logistic curves
PH curves

Note: $y = (x + \frac{1}{x})^2 + 5(x + \frac{1}{x}) + 6.25$
 $= x^2 + \frac{1}{x^2} + 5x + \frac{5}{x} + 6.25$